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Title: Management of Hypocalcemia

Symptomatic Hypocalcemia

- Carpopedal spasm
- Convulsion
- Jitteriness
- Apnea

1. Start Intravenous 10% calcium gluconate (1 ml = 9 mg of elemental calcium)

- Bolus dose 2 ml/kg – over 10-30 minutes
- Should be under cardiac monitor
- Dose can be repeated if necessary

2. Maintenance calcium infusion at a rate of 4 ml / kg / day of 10% calcium gluconate with 1:5 dilution with 5% dextrose.

- Monitor calcium 4 - 6 hours

Note:

1. Sudden cessation of IV calcium can cause rebound hypocalcemia

2. The infusion should be titrated according to serum calcium level. Once the adjusted calcium above 2.1 mmol/l infusion should be tapered by 0.5 ml/kg/day

4. Vitamin D3 / Alpha calcidol should be started while the patient is on calcium infusion.

5. The following blood investigations should be done:

- PTH, Magnesium, Vitamin D (before starting Vitamin D).

Important Message:

1. Bicarbonate must not be infused in the same infusion line.

2. Easily flowing IV line should be secured since extravasated calcium can cause serious tissue necrosis.

Asymptomatic Hypocalcemia

1. Oral calcium at a dose of:

- 50-100 mg elemental calcium/kg/day

2. Vitamin D (alfacalcidol) should be given when S. calcium reaches 2 mmol/L

* Doses of vitamin D should be given according to the underlying pathogenesis

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Pediatric Endocrine textbook, 4 th edition	Sperling	2008	745

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Review Details

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